



WeCloudData

LangChain

Introduction





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Introduction LangChain

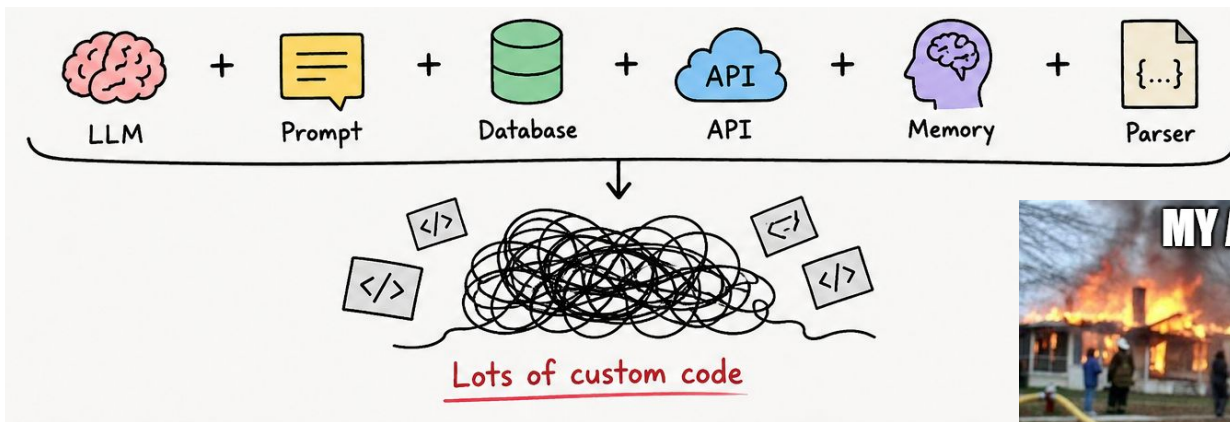


Why Do We Need LangChain?

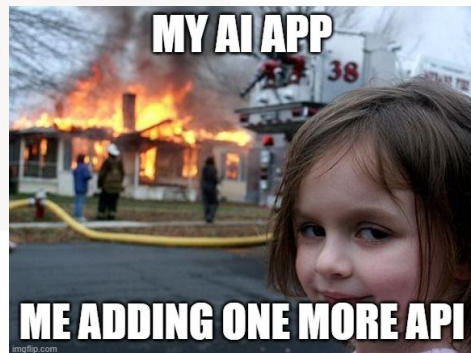
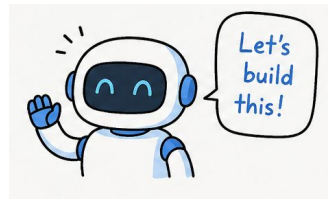
LangChain

Imagine building an AI Assistant:

Without LangChain, you have to manually connect:



Your code quickly becomes difficult to manage.





What is LangChain?

LangChain

LangChain is an **open-source framework** that helps developers build applications powered by Large Language Models.

Instead of writing everything from scratch, LangChain provides reusable building blocks for creating AI workflows.



LangChain

<https://techcommunity.microsoft.com/blog/azuredevcommunityblog/langchain-v1-is-now-generally-available/4462159>



Why and When to use LangChain?

LangChain

Why?

- Reduce development time
- Connect LLMs with external data and tools
- Build complex AI applications more easily
- Organize AI workflows into reusable components

When?

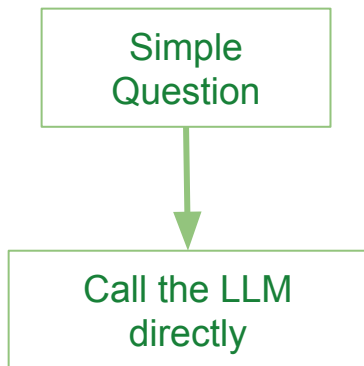
Use LangChain when your application needs more than just a single prompt, such as:

- Chatbots
- Document Q&A
- AI Agents
- RAG systems
- Multi-step workflows

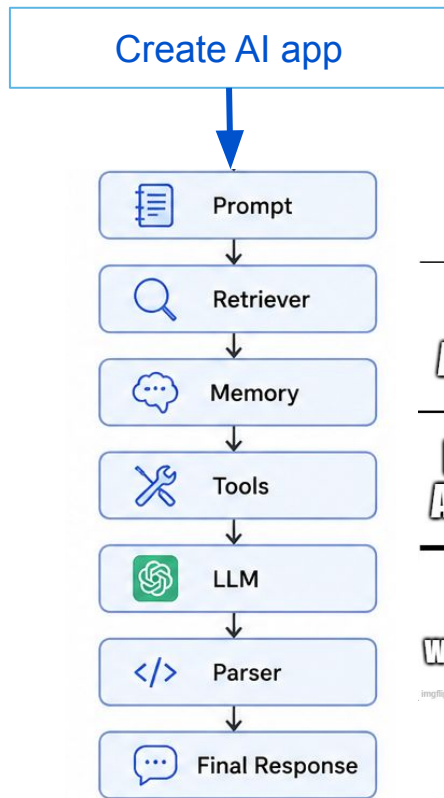


When do we actually need LangChain?

LangChain



LangChain becomes valuable when your application involves **multiple components working together**.



CALLING
AN LLM

ADDING
A PROMPT

BUILDING
A CHATBOT

BUILDING
AN AI AGENT
WITH LANGCHAIN





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LangChain Building Blocks

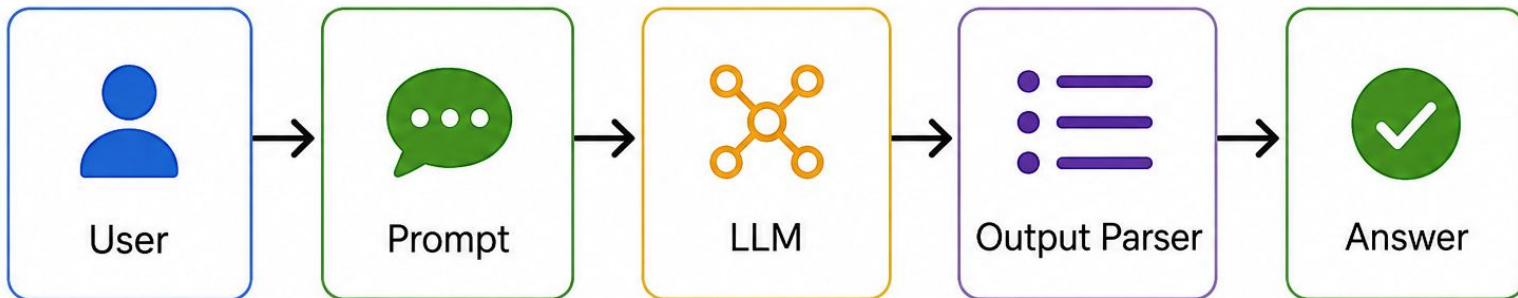
LangChain



The Big Picture

LangChain

Everything in LangChain is built from reusable components.



Later we will add more components.



Prompts

LangChain

Prompts tell the LLM **what to do** and **how to respond**.

Why?

- Reusable
- Dynamic
- Easy to modify



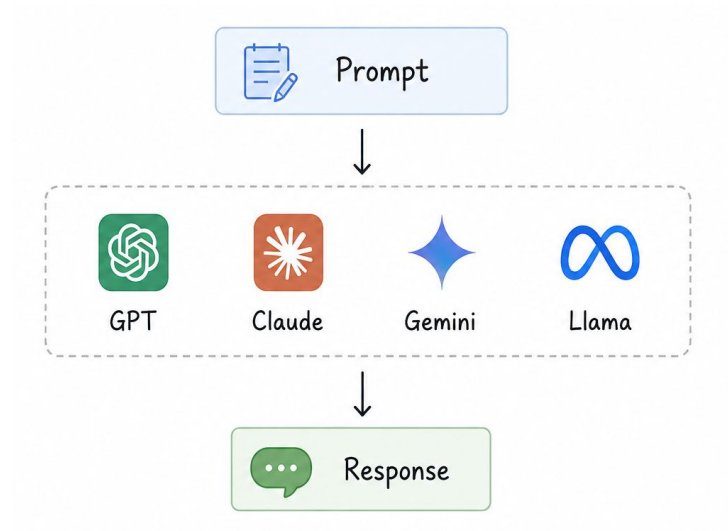


Models

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Models are the brains of your AI application.

LangChain lets you switch models without changing your entire application.





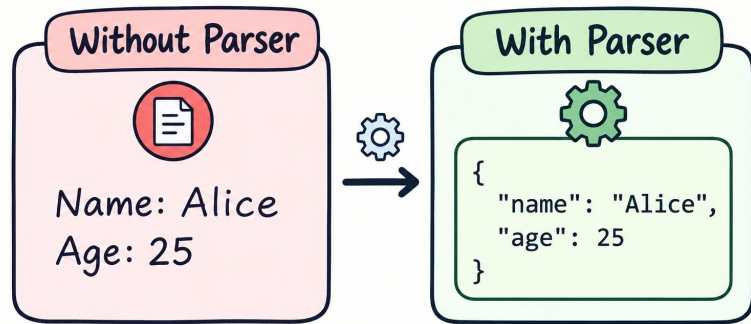
Output Parsers

LangChain

LLMs usually return plain text.

Sometimes we need

- JSON
- Lists
- Tables
- Structured output



Output parsers convert raw responses into a predictable format.



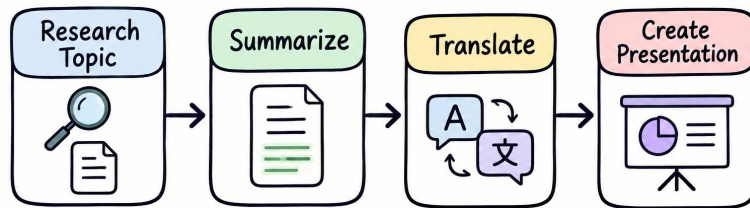
Chains

LangChain

A Chain connects multiple steps together.

Instead of one prompt, the output of one step becomes the input of the next.

Chains in LangChain



Why?

Break large tasks into manageable steps.

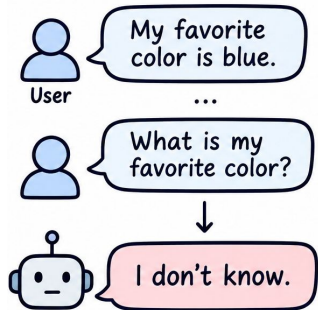


Memory

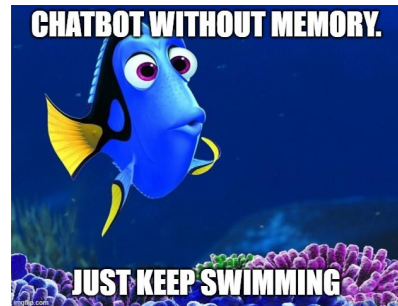
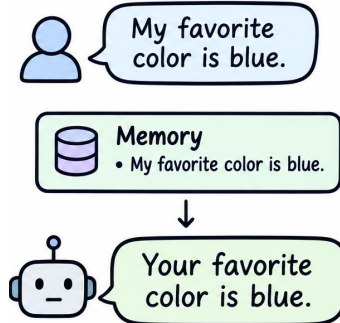
LangChain

Memory allows AI applications to remember previous conversations.

Without Memory



With memory





Retrievers

LangChain

What?

Retrievers search for relevant documents before the LLM answers.

Where?

Used in

- RAG
- Enterprise search
- Document chatbots



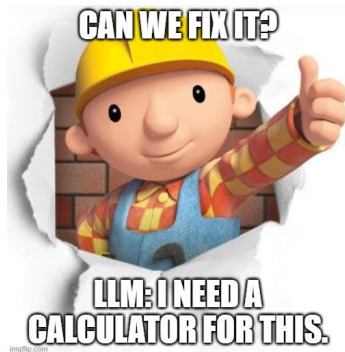
Tools

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Tools allow an LLM to interact with external systems.

Examples

- Calculator
- Weather API
- Search Engine
- Database
- Email





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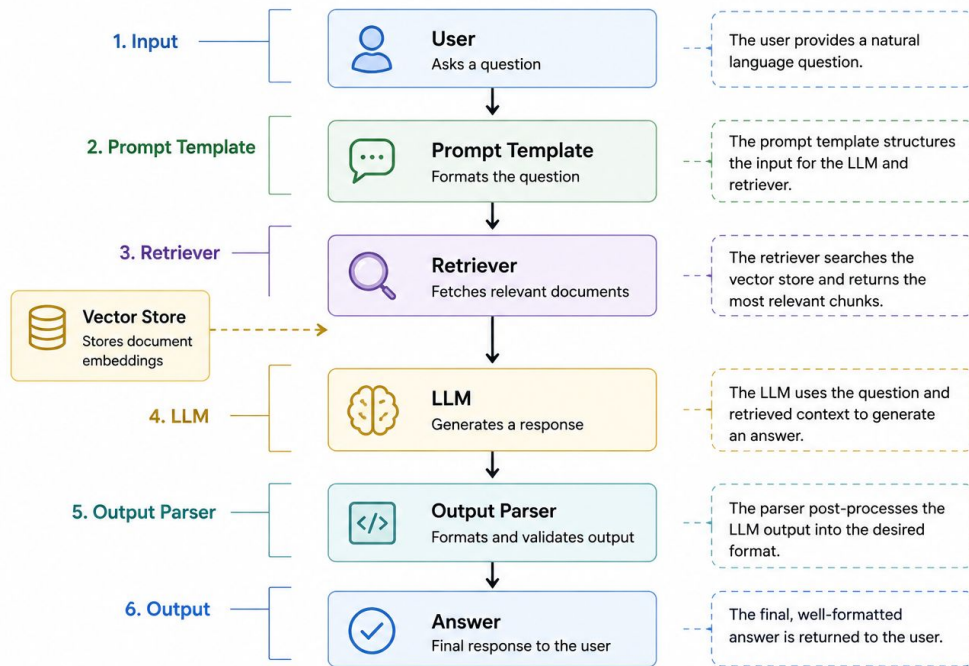
Putting It All Together LangChain



How Components Work Together

LangChain

LangChain Components: End-to-End Flow

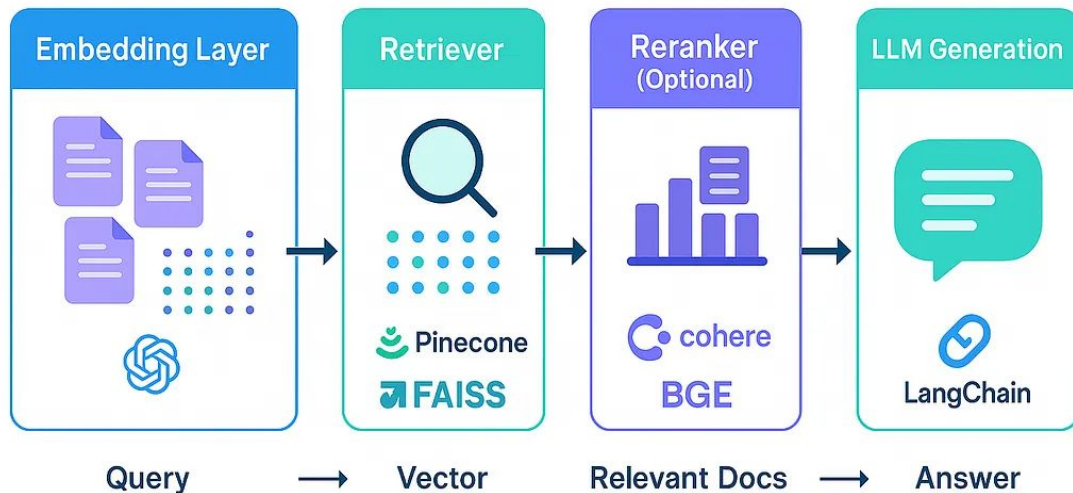




Building a RAG Application

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LangChain RAG Architecture



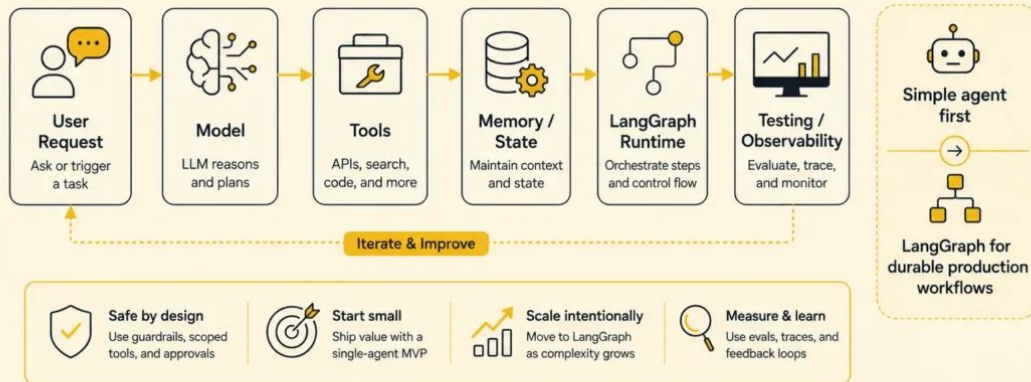


Building an AI Agent

LangChain

Build AI Agents with LangChain

Start small, use safe tools, and scale with LangGraph when workflows grow.



www.designveloper.com

<https://www.designveloper.com/blog/how-to-build-ai-agents-with-langchain/>



Why Developers Like LangChain?

LangChain

Without LangChain

- Lots of custom code
- Difficult maintenance
- Hard to extend
- More debugging

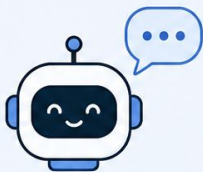
With LangChain

- Reusable components
- Modular workflows
- Easy to add new components
- Faster development



Common LangChain Applications

LangChain



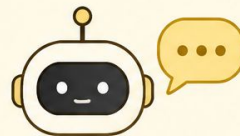
Chatbots



Customer Support



Document Q&A



AI Assistants



Code Assistants



Research Assistants



AI Agents



Summary

- What LangChain is
- Why it exists
- When to use it
- Core building blocks
- How components work together
- Real-world AI applications



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!